Choose axes that help explain the ramp

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Weight components / ES: componentes del peso / PT: componentes do peso

mg sin θ along; mg cos θ into the ramp.

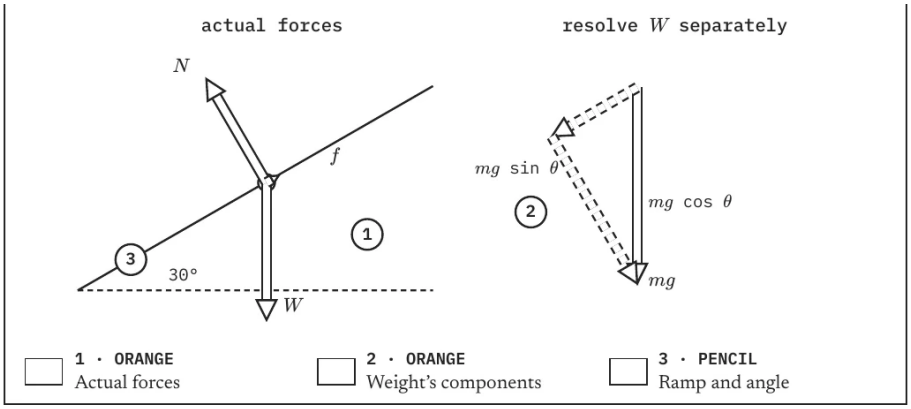
Also on your note page: Parallel axis, Perpendicular axis, Constraint.

B READ THE RELATIONSHIP

Downhill: ma = mg sin θ − f. Perpendicular: N = mg cos θ here.

As you read, find: parallel · perpendicular · downhill.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 2 kg block slides down a 30° ramp with 2 N friction uphill. Find its acceleration and normal force. Use g = 9.8 m/s² .

Given m = 2 kg ; θ = 30° f = 2 N uphill a ⊥ = 0

Rule mg sin θ − f = ma ∥ N = mg cos θ

Work a ∥ = [19.6(0.5) − 2]/2 N = 19.6 cos 30°

Answer 3.9 m/s² Downhill; N ≈ 17.0 N .

E SAVE FOR THE FINAL MEETING

A suitable set of axes can simplify the forces changing XJ's path. Components are calculation tools; they must not be counted again as extra forces.

Read the physics: Weight components

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

On a ramp, gravity still points straight down. The normal force points straight out of the surface. Choose one axis along the ramp and one perpendicular to it.

02 HOW IT WORKS

Split weight into two parts. The parallel part pulls downhill; the perpendicular part presses into the ramp. Calculate with the parts or with the original weight, never with both together.

03 WHAT TO CHECK

Subtract uphill friction from the downhill force component. If the block stays on the ramp, its acceleration perpendicular to the ramp is zero.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 2 kg block slides down a 30° ramp with 2 N friction uphill. Find its acceleration and normal force. Use $g = 9.8\ m/s^2$. $3.9\ m/s^2$ Downhill; $N \approx 17.0\ N$.

Model check: Downhill: $ma = mg \sin\theta - f$. Perpendicular: $N = mg \cos\theta$ here.

05 CONNECT TO XJ

A suitable set of axes can simplify the forces changing XJ's path. Components are calculation tools; they must not be counted again as extra forces.

SAY IT BEFORE YOU WRITE

Why would adding both weight and its components be a mistake? Along the ramp I use ___. Across the ramp I use ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Weight components / ES: componentes del peso / PT: componentes do peso

Say it in any language. Draw or use a few words. Use the notes to check your idea.

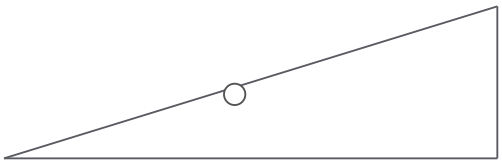
Meaning	How can you tell?
To me, weight components means …	Which part of the weight pulls downhill? Which part presses into the ramp? One part points ___. The other points ___.
Example or sketch	**Non-example**
I see it when …	This looks close, but …

B DRAW WITH YOUR TEACHER

Draw the real weight once, then draw its two components in a separate box. Circle the component along the ramp.

Labels you can use: parallel · perpendicular · downhill.

Real forces



Components

Tell a partner: Along the ramp I use ___. Across the ramp I use ___.

On assignment: Weight components

SECTION E / BACK / FINAL MEETING ONLY Name _____

A navigation graphic lists a gravitational force AND its two components as three separate pulls on XJ.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Navigation engineer**

Build an overlay using the component view. Indicate the full arrow it replaces before releasing the values to the calculator.

☐ **Map graphics editor**

Sketch two camera views: the real pull, then its components. Give viewers a caption that prevents counting the same pull twice.

Word bank: parallel · perpendicular · downhill.

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

“We first said ___. Now we ___ because ___.”

Editor’s check: State which component can change speed at that instant.